

Ishaan Pathak

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EDUCATION

Lawrence Technological University
Master's of Science in Computer Science
Teaching Assistant: Python & Robotics

Devi Ahilya VishwaVidyalaya
Bachelor of Computer Application (Honours)

Southfield, MI
August 2024 – July 2026

Indore, India
August 2019 – July 2023

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS
AI/ML: PyTorch, TensorFlow, Keras, Transformers (Hugging Face), Computer Vision, RNNs, CNNs, YOLO, scikit-learn, scipy
Generative AI: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, Semantic Search
Web Development: React.js, Node.js, Express.js, Flask, Django, Spring Boot, Three.js, REST APIs
Cloud & DevOps: AWS (Lambda, DynamoDB, S3 Buckets), Docker, Kubernetes, CI/CD, GitHub Actions, Jenkins, Kafka
Databases: MySQL, PostgreSQL, MongoDB, Oracle, Redis, NoSQL, GraphQL

EXPERIENCE

Research Assistant February 2025 – July 2026
Lawrence Technological University Southfield, MI

- **Achieved 81%** mAP on YOLO-based crack detection models by curating and **annotating a 10,000+** image dataset
- **Developed** state-of-the-art Computer Vision models for medical detection, contributing to automated medical research.
- **Assisted** professor in **reviewing, refining, publishing** research manuscripts for clarity, rigor, and publication readiness.

Student Web Developer May 2025 – August 2025
Lawrence Technological University Southfield, MI

- **Developed** Web app for anonymous essay submissions, enabling over **50+ students** to participate in bias-free evaluations.
- **Integrated** a Gen AI model to benchmark automated grading against human assessments, **improving** consistency by **25%**.
- **Optimized** the grading workflow, **reducing** manual evaluation time by **40%** and grading throughput **increased by 35%**.

Junior AI & ML Engineer July 2023 – July 2024
Essar Agro Indore, India

- **Cleaned** and standardized **1M+** procurement records (JSON, Excel, CSV) via a unified ingestion pipeline.
- **Automated** livestock medication dosing by **developing** LSTM/GRU models on real-time client text data.
- **Built** CI/CD workflows using Docker and GitHub Actions, self-taught through hands-on testing.

Junior Machine Learning Engineer June 2022 – February 2023
GenieTalk.ai Indore, India

- **Developed** and **optimized** machine learning models for text classification using TensorFlow and scikit-learn.
- **Enhanced** algorithm performance and **integrated** AWS for scalable model deployment.
- **Developed** Application Programming Interface (APIs) for model integration, enhancing response times by **20%**.

PROJECTS

MedBot360 – AI Health Companion | *MERN, Transformers (Hugging Face), Docker, AWS*

- **Built** a MERN-based health dashboard enabling secure symptom tracking and personalized recommendations for **200+** users.
- **Integrated** a Hugging Face-powered medical chatbot for real-time symptom analysis, boosting accuracy by 30%
- Containerized the full stack using **Docker** and deployed on **AWS** for high availability and scalability.

Social Media Analysis | *Apache Spark, Scikit-Learn, Python (Matplotlib, Seaborn, Pandas), SQL*

- **Engineered** an **Apache Spark** pipeline to process high-volume tweet data using distributed computing.
- **Implemented** **K-Means** and **PCA** to automate content segmentation and identify sentiment trends.
- **Automated** visual reporting via **Matplotlib/Seaborn** to deliver actionable insights for brand monitoring.

Personalized Search Reranker | *BERT, Cross-Encoders, TREC DL, PyTorch*

- **Boosted** search accuracy (**MRR@10 to 0.0406**) by developing a reranking pipeline integrated with clickstream data.
- Predicted **query-document relevance** by fine-tuning Sentence-BERT on MS MARCO using binary cross-entropy.
- Enhanced **inference efficiency by 15%** by scaling Python preprocessing and optimizing **AdamW** training.

RESEARCH, AWARDS & CERTIFICATIONS

- Publications: **AI-Powered Music Generation from Sequential Motion Signals** (IJIIS 2025); **Automated Brain Tumor Detection via YOLOv11** (JCTR 2025)
- Research: **Smart Infrastructure Damage Detection** using Computer Vision for structural health monitoring.
- **Outstanding Research Assistant in Data Science Award** — Lawrence Technological University